Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Conversational AI, Agentic AI, Prompt Engineering, Automated QA Testing, Data Governance, Privacy & Compliance, Dimensional Modeling (Star/Snowflake Schemas, SCD Type 1/2), Predictive Modeling & Machine Learning (XGBoost, Random Forests, SARIMA, Scikit-Learn), Automated Data Quality Audits, High-Throughput ETL, Root Cause Analysis, Capacity Planning, Self-Serve Analytics Tooling

Programming Languages & Tools: GitHub Copilot, KIRO IDE, Azure Data Factory (ADF), Azure Synapse, Azure SQL, Databricks, AWS (S3, Redshift), Snowflake, Apache Spark, PySpark, Python, SQL, Power BI (DAX), Tableau, MicroStrategy (MSTR), Airflow, One Automation

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior AI Analytics & Tooling Engineer — **May 2022 – Present**

- **Architected conversational AI tools and automated** QA testing frameworks (SpectrumIQ) using prompt engineering and Agentic AI workflows, cutting manual turnaround time by 50% and building self-serve capabilities for analysts.
- **Engineered automated data quality audit frameworks** with firewalls and high-performance SQL stored procedures, reducing operational turnaround by over 80% while ensuring strict adherence to data governance and privacy requirements.
- **Leveraged Agentic AI and IDE-assisted development** such as KIRO IDE for automated profiling and regex classification, accelerating feature delivery by 40% and building scalable data tooling.
- **Modeled complex dimensional data schemas** (Star/Snowflake) and implemented Slowly Changing Dimensions (SCD Type 1/2) for high-throughput data pipelines, ensuring trusted and consistent data availability (99.9% SLA).
- **Deployed proactive anomaly detection and predictive** machine learning models with 95% accuracy using Python, XGBoost, and Scikit-Learn to mitigate ETL discrepancies and optimize workforce capacity planning.

Graduate Research Data Scientist — **May 2022 – Present**

- **Developed advanced predictive models and statistical** forecasting pipelines using Python and Scikit-Learn to analyze complex datasets and translate empirical insights into scalable tools.
- **Applied rigorous data science methodologies** to evaluate operational metrics, driving data-informed process improvements and cross-functional research collaboration.
- **Synthesized large-scale datasets into actionable analytics** dashboards using advanced BI tools, empowering stakeholders with self-serve insights.

Petrochemical Data Systems Engineer — **May 2022 – Present**

- **Designed data collection systems and analytical** workflows to monitor large-scale industrial processes and optimize operational efficiency.
- **Automated reporting pipelines and data audits** to minimize error rates and enforce strict compliance with engineering and quality standards.
- **Collaborated across multidisciplinary teams** to translate operational requirements into robust data-driven tooling and technical documentation.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.